Lab 6: CUDA Sobel Edge Detection

EEL6763 – Parallel Computer Architecture

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Part 1: CUDA Sobel Edge Detection Using Global Memory

1.1 Kernel Code

Kernel 1: 1D Global Memory

```
__global__ void sobel_1d(const uint8_t * __restrict__ in, uint8_t * __restrict__ out, int width, int height) {
  int idx = blockDim.x * blockIdx.x + threadIdx.x;

  if (idx >= width * height) return;

  int r = idx / width;
  int c = idx % width;

  if (r == 0 || r == height - 1 || c == 0 || c == width - 1) {
    out[idx] = 0;
    return;
  }

  int p1 = in[(r-1)*width + (c-1)];
  int p2 = in[(r-1)*width +  c];
  int p3 = in[(r-1)*width + (c+1)];
  int q1 = in[r*width + (c-1)];
  int q3 = in[r*width + (c+1)];
  int r1 = in[(r+1)*width + (c-1)];
  int r2 = in[(r+1)*width + c];
```

```
    int r3 = in[(r+1)*width + (c+1)];

    int qx = (p1 - r1) + 2*(p2 - r2) + (p3 - r3);
    int qy = (p1 - p3) + 2*(q1 - q3) + (r1 - r3);
    int mag = abs(qx) + abs(qy);
    if (mag > 255) mag = 255;

    out[idx] = (uint8_t)mag;
}
```

Kernel 2: 2D Global Memory

```
__global__ void sobel_2d(const uint8_t * __restrict__ in, uint8_t * __restrict__ out, int width, int height) {
    /* Global thread index – 2-D cheatsheet formula */
    int c = blockDim.x * blockIdx.x + threadIdx.x;
    int r = blockDim.y * blockIdx.y + threadIdx.y;

    if (r >= height || c >= width) return;

    /* Skip border pixels */
    if (r == 0 || r == height - 1 || c == 0 || c == width - 1) {
        out[r*width + c] = 0;
        return;
    }

    int p1 = in[(r-1)*width + (c-1)];
    int p2 = in[(r-1)*width +  c];
    int p3 = in[(r-1)*width + (c+1)];
    int q1 = in[r*width + (c-1)];
    int q3 = in[r*width + (c+1)];
```

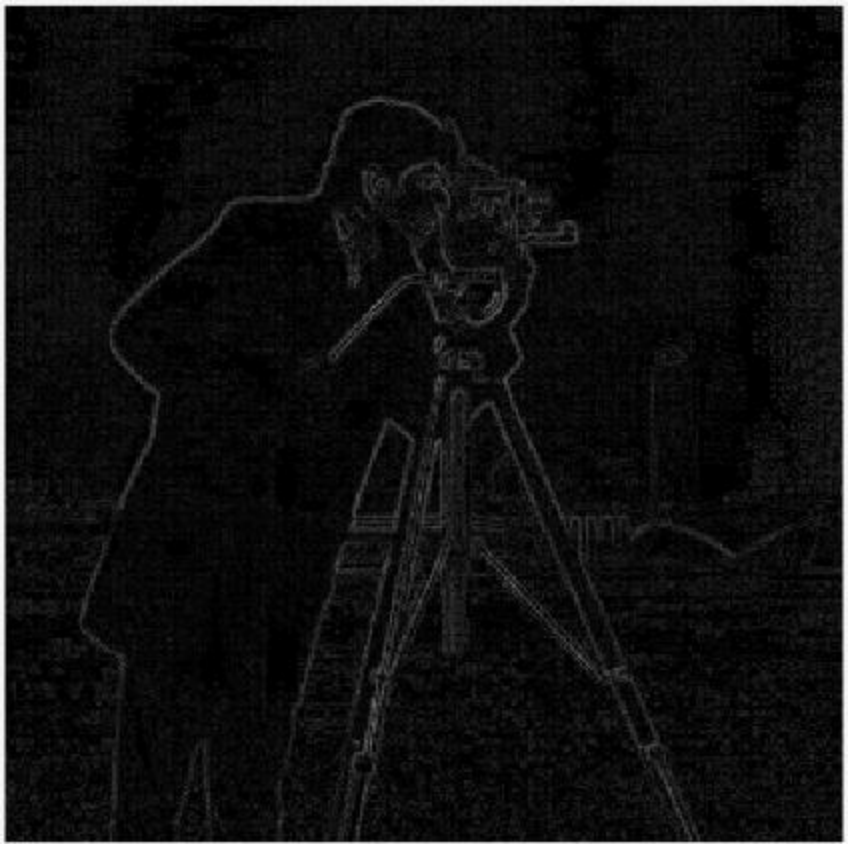
```
    int r1 = in[(r+1)*width + (c-1)];
    int r2 = in[(r+1)*width + c];
    int r3 = in[(r+1)*width + (c+1)];

    int qx = (p1 - r1) + 2*(p2 - r2) + (p3 - r3);
    int qy = (p1 - p3) + 2*(q1 - q3) + (r1 - r3);
    int mag = abs(qx) + abs(qy);
    if (mag > 255) mag = 255;

    out[r*width + c] = (uint8_t)mag;
}
```

1.2 Part 1 Output Images/ Execution Time

Configuration 1 Execution:

```
==PROF== Disconnected from process 935297
[935297] sobel_cuda@127.0.0.1
  sobel_1d(const unsigned char *, unsigned char *, int, int) (25000, 1, 1)x(1000, 1, 1), Device 0, CC 8.9, Invocations 1
    Section: GPU Speed Of Light Throughput
    ----------------------- ----------- ---------- ---------- ----------
    Metric Name             Metric Unit    Minimum    Maximum    Average
    ----------------------- ----------- ---------- ---------- ----------
    DRAM Frequency                  Ghz       6.24       6.24       6.24
    SM Frequency                    Mhz     827.53     827.53     827.53
    Elapsed Cycles                cycle 810,283.00 810,283.00 810,283.00
    Memory Throughput                 %      34.54      34.54      34.54
    DRAM Throughput                   %      13.33      13.33      13.33
    Duration                         us     972.64     972.64     972.64
    L1/TEX Cache Throughput           %      38.87      38.87      38.87
    L2 Cache Throughput               %      15.05      15.05      15.05
    SM Active Cycles              cycle 710,440.19 710,440.19 710,440.19
    Compute (SM) Throughput           %      34.54      34.54      34.54
    ----------------------- ----------- ---------- ---------- ----------

    Section: GPU and Memory Workload Distribution
    ------------------------- ----------- -------------- -------------- --------------
    Metric Name               Metric Unit        Minimum        Maximum        Average
    ------------------------- ----------- -------------- -------------- --------------
    Average DRAM Active Cycles       cycle     809,866.67     809,866.67     809,866.67
    Total DRAM Elapsed Cycles        cycle  36,440,064.00  36,440,064.00  36,440,064.00
    Average L1 Active Cycles         cycle     710,440.19     710,440.19     710,440.19
    Total L1 Elapsed Cycles          cycle  46,366,622.00  46,366,622.00  46,366,622.00
    Average L2 Active Cycles         cycle     790,625.50     790,625.50     790,625.50
    Total L2 Elapsed Cycles          cycle  19,632,456.00  19,632,456.00  19,632,456.00
    Average SM Active Cycles         cycle     710,440.19     710,440.19     710,440.19
    Total SM Elapsed Cycles          cycle  46,366,622.00  46,366,622.00  46,366,622.00
    Average SMSP Active Cycles       cycle     696,487.57     696,487.57     696,487.57
    Total SMSP Elapsed Cycles        cycle 185,466,488.00 185,466,488.00 185,466,488.00
    ------------------------- ----------- -------------- -------------- --------------

    Section: Launch Statistics
    -------------------------------- --------------- ------------- ------------- -------------
    Metric Name                          Metric Unit       Minimum       Maximum       Average
    -------------------------------- --------------- ------------- ------------- -------------
    Block Size                                             1,000.00      1,000.00      1,000.00
    Grid Size                                             25,000.00     25,000.00     25,000.00
    Registers Per Thread             register/thread          20.00         20.00         20.00
    Shared Memory Configuration Size           Kbyte           8.19          8.19          8.19
    Driver Shared Memory Per Block       Kbyte/block           1.02          1.02          1.02
    Dynamic Shared Memory Per Block       byte/block           0.00          0.00          0.00
    Static Shared Memory Per Block        byte/block           0.00          0.00          0.00
    # SMs                                          SM          58.00         58.00         58.00
    Stack Size                                            1,024.00      1,024.00      1,024.00
    Threads                                  thread 25,000,000.00 25,000,000.00 25,000,000.00
    # TPCs                                                 29.00         29.00         29.00
    Uses Green Context                                      0.00          0.00          0.00
    Waves Per SM                                          431.03        431.03        431.03
    -------------------------------- --------------- ------------- ------------- -------------

    Section: Occupancy
    ------------------------------ ----------- ------- ------- -------
    Metric Name                    Metric Unit Minimum Maximum Average
    ------------------------------ ----------- ------- ------- -------
    Block Limit SM                       block   24.00   24.00   24.00
    Block Limit Registers                block    2.00    2.00    2.00
    Block Limit Shared Mem               block    8.00    8.00    8.00
    Block Limit Warps                    block    1.00    1.00    1.00
    Theoretical Active Warps per SM       warp   32.00   32.00   32.00
    Theoretical Occupancy                    %   66.67   66.67   66.67
    Achieved Occupancy                       %   60.39   60.39   60.39
    Achieved Active Warps Per SM          warp   28.99   28.99   28.99
    ------------------------------ ----------- ------- ------- -------

  Note: The shown averages are calculated as the arithmetic mean of the metric values after the evaluation of the
  metrics for each individual kernel launch.
  If aggregating across varying launch configurations (like shared memory, cache config settings), the arithmetic
  mean can be misleading and looking at the individual results is recommended instead.
  This output mode is backwards compatible to the per-kernel summary output of nvprof


Total program wall-clock time: 1.81528032 sec

[tbarber@login8 Part1]$
```

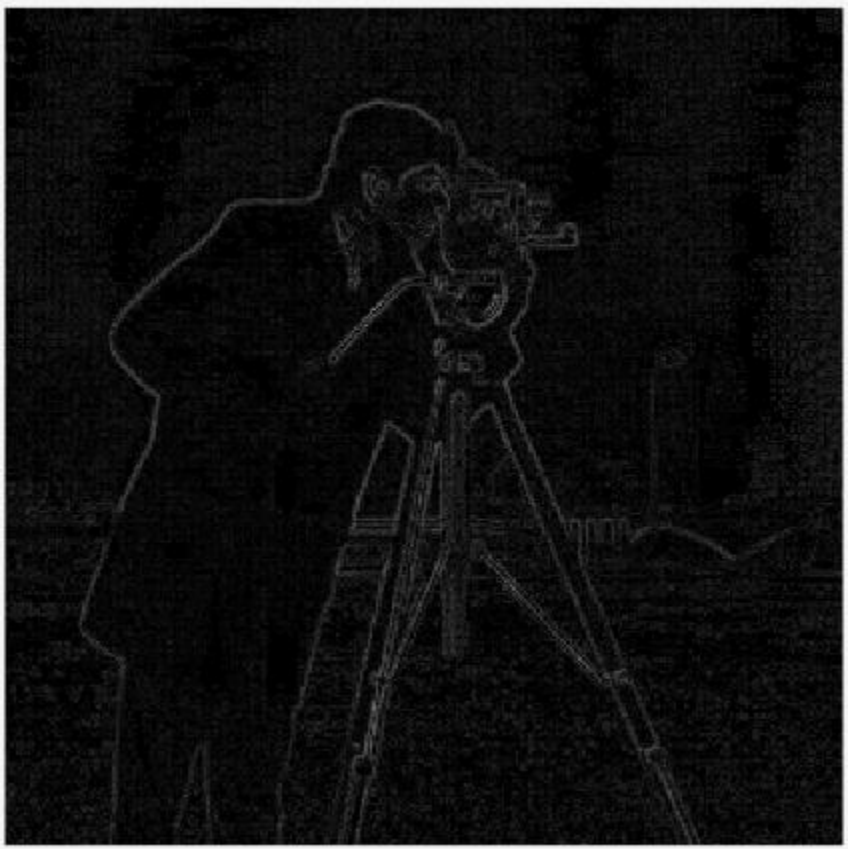
Configuration 1 Output:



Configuration 2 Execution:

```
==PROF== Disconnected from process 1956257
[1956257] sobel_cuda@127.0.0.1
  sobel_2d(const unsigned char *, unsigned char *, int, int) (50, 500, 1)x(100, 10, 1), Device 0, CC 8.9, Invocations 1
    Section: GPU Speed Of Light Throughput
    ----------------------- ----------- ---------- ---------- ----------
    Metric Name             Metric Unit    Minimum    Maximum    Average
    ----------------------- ----------- ---------- ---------- ----------
    DRAM Frequency                  Ghz       6.24       6.24       6.24
    SM Frequency                    Mhz     811.90     811.90     811.90
    Elapsed Cycles                cycle 652,383.00 652,383.00 652,383.00
    Memory Throughput                 %      47.19      47.19      47.19
    DRAM Throughput                   %      16.44      16.44      16.44
    Duration                         us     794.72     794.72     794.72
    L1/TEX Cache Throughput           %      54.89      54.89      54.89
    L2 Cache Throughput               %      19.51      19.51      19.51
    SM Active Cycles              cycle 553,340.21 553,340.21 553,340.21
    Compute (SM) Throughput           %      47.19      47.19      47.19
    ----------------------- ----------- ---------- ---------- ----------

    Section: GPU and Memory Workload Distribution
    ------------------------- ----------- -------------- -------------- --------------
    Metric Name               Metric Unit        Minimum        Maximum        Average
    ------------------------- ----------- -------------- -------------- --------------
    Average DRAM Active Cycles       cycle     815,741.33     815,741.33     815,741.33
    Total DRAM Elapsed Cycles        cycle  29,773,824.00  29,773,824.00  29,773,824.00
    Average L1 Active Cycles         cycle     553,340.21     553,340.21     553,340.21
    Total L1 Elapsed Cycles          cycle  37,325,718.00  37,325,718.00  37,325,718.00
    Average L2 Active Cycles         cycle     611,735.12     611,735.12     611,735.12
    Total L2 Elapsed Cycles          cycle  15,734,112.00  15,734,112.00  15,734,112.00
    Average SM Active Cycles         cycle     553,340.21     553,340.21     553,340.21
    Total SM Elapsed Cycles          cycle  37,325,718.00  37,325,718.00  37,325,718.00
    Average SMSP Active Cycles       cycle     541,200.94     541,200.94     541,200.94
    Total SMSP Elapsed Cycles        cycle 149,302,872.00 149,302,872.00 149,302,872.00
    ------------------------- ----------- -------------- -------------- --------------

    Section: Launch Statistics
    -------------------------------- --------------- ------------- ------------- -------------
    Metric Name                          Metric Unit       Minimum       Maximum       Average
    -------------------------------- --------------- ------------- ------------- -------------
    Block Size                                             1,000.00      1,000.00      1,000.00
    Grid Size                                             25,000.00     25,000.00     25,000.00
    Registers Per Thread             register/thread          20.00         20.00         20.00
    Shared Memory Configuration Size           Kbyte           8.19          8.19          8.19
    Driver Shared Memory Per Block       Kbyte/block           1.02          1.02          1.02
    Dynamic Shared Memory Per Block       byte/block           0.00          0.00          0.00
    Static Shared Memory Per Block        byte/block           0.00          0.00          0.00
    # SMs                                         SM          58.00         58.00         58.00
    Stack Size                                            1,024.00      1,024.00      1,024.00
    Threads                                  thread 25,000,000.00 25,000,000.00 25,000,000.00
    # TPCs                                               29.00         29.00         29.00
    Uses Green Context                                     0.00          0.00          0.00
    Waves Per SM                                         431.03        431.03        431.03
    -------------------------------- --------------- ------------- ------------- -------------

    Section: Occupancy
    ------------------------------- ----------- ------- ------- -------
    Metric Name                     Metric Unit Minimum Maximum Average
    ------------------------------- ----------- ------- ------- -------
    Block Limit SM                        block   24.00   24.00   24.00
    Block Limit Registers                 block    2.00    2.00    2.00
    Block Limit Shared Mem                block    8.00    8.00    8.00
    Block Limit Warps                     block    1.00    1.00    1.00
    Theoretical Active Warps per SM        warp   32.00   32.00   32.00
    Theoretical Occupancy                     %   66.67   66.67   66.67
    Achieved Occupancy                        %   57.04   57.04   57.04
    Achieved Active Warps Per SM           warp   27.38   27.38   27.38
    ------------------------------- ----------- ------- ------- -------

  Note: The shown averages are calculated as the arithmetic mean of the metric values after the evaluation of the
  metrics for each individual kernel launch.
  If aggregating across varying launch configurations (like shared memory, cache config settings), the arithmetic
  mean can be misleading and looking at the individual results is recommended instead.
  This output mode is backwards compatible to the per-kernel summary output of nvprof


Total program wall-clock time: 2.05266304 sec

[tbarber@login8 Part1]$
```

Configuration 2:



1.3 Performance Table

Grid Configuration	Block Configuration	Kernel Time (s)	Wall Time (s)
(25000,1,1)	(1000,1,1)	0.000973	1.815
(50,500,1)	(100,10,1)	0.000795	2.053

1.4 Comparison with Lab 4

Implementation	Execution Time (s)
CPU	2.037
CPU + MPI	1.014
GPU (Global Memory)	1.815

1.5 Analysis

The CUDA implementation using global memory demonstrated improved performance compared to the serial CPU implementation, reducing execution time from approximately 2.037 seconds to 1.815 seconds. However, the MPI implementation achieved the best performance at approximately 1.014 seconds.

While Configuration 2 achieved a lower kernel execution time, Configuration 1 produced the lowest overall wall clock time, indicating that total execution time is influenced not only by kernel computation but also by memory transfer and execution overhead.

Kernel execution time was significantly smaller than total wall time, demonstrating that host-device memory transfers contributed substantially to total execution time.

The best configuration for global memory was:

Grid = (25000,1,1), Block = (1000,1,1)

Part 2: CUDA Sobel Edge Detection Using Shared Memory

2.1 Kernel Code

Kernel 1: 1D Shared Memory

```
__global__ void sobel_1d_shared(const uint8_t * __restrict__ in, uint8_t * __restrict__ out, int width, int height) {
  int tid = blockDim.x * blockIdx.x + threadIdx.x;
  int r = tid / width;
  int c = tid % width;
  int tx = threadIdx.x;
  int bw = blockDim.x;
  int smem_w = bw + 2;

  extern __shared__ uint8_t smem[];

  smem[0 * smem_w + (tx + 1)] = (r - 1 >= 0 && c >= 0 && c < width) ? in[(r-1)*width + c] : 0;
  smem[1 * smem_w + (tx + 1)] = (r >= 0 && r < height && c >= 0 && c < width) ? in[r*width + c] : 0;
  smem[2 * smem_w + (tx + 1)] = (r + 1 < height && c >= 0 && c < width) ? in[(r+1)*width + c] : 0;

  if (tx == 0) {
```

```
    smem[0 * smem_w + 0] = (r - 1 >= 0 && c - 1 >= 0) ? in[(r-1)*width + (c-1)] : 0;
    smem[1 * smem_w + 0] = (r >= 0 && r < height && c - 1 >= 0) ? in[r*width + (c-1)] : 0;
    smem[2 * smem_w + 0] = (r + 1 < height && c - 1 >= 0) ? in[(r+1)*width + (c-1)] : 0;
}

if (tx == bw - 1) {
    smem[0 * smem_w + (bw + 1)] = (r - 1 >= 0 && c + 1 < width) ? in[(r-1)*width + (c+1)] : 0;
    smem[1 * smem_w + (bw + 1)] = (r >= 0 && r < height && c + 1 < width) ? in[r*width + (c+1)] : 0;
    smem[2 * smem_w + (bw + 1)] = (r + 1 < height && c + 1 < width) ? in[(r+1)*width + (c+1)] : 0;
}

__syncthreads();

if (tid >= width * height) return;
if (r == 0 || r == height - 1 || c == 0 || c == width - 1) {
    out[tid] = 0;
    return;
}

int sc = tx + 1;
int p1 = smem[0 * smem_w + (sc - 1)];
int p2 = smem[0 * smem_w + sc];
int p3 = smem[0 * smem_w + (sc + 1)];
int q1 = smem[1 * smem_w + (sc - 1)];
int q3 = smem[1 * smem_w + (sc + 1)];
int r1 = smem[2 * smem_w + (sc - 1)];
int r2 = smem[2 * smem_w + sc];
int r3 = smem[2 * smem_w + (sc + 1)];
```

```
    int qx = (p1 - r1) + 2*(p2 - r2) + (p3 - r3);
    int qy = (p1 - p3) + 2*(q1 - q3) + (r1 - r3);
    int mag = abs(qx) + abs(qy);
    if (mag > 255) mag = 255;

    out[tid] = (uint8_t)mag;
}
```

Kernel 2: 2D Shared Memory

```
__global__ void sobel_2d_shared(const uint8_t * __restrict__ in, uint8_t * __restrict__ out, int width, int height) {
    int c = blockDim.x * blockIdx.x + threadIdx.x;
    int r = blockDim.y * blockIdx.y + threadIdx.y;
    int tx = threadIdx.x;
    int ty = threadIdx.y;
    int bw = blockDim.x;
    int bh = blockDim.y;
    int smem_w = bw + 2;

    extern __shared__ uint8_t smem[];

    smem[(ty + 1) * smem_w + (tx + 1)] = (r >= 0 && r < height && c >= 0 && c < width) ? in[r*width + c] : 0;

    if (ty == 0)
        smem[0 * smem_w + (tx + 1)] = (r - 1 >= 0 && c >= 0 && c < width) ? in[(r-1)*width + c] : 0;
    if (ty == bh - 1)
        smem[(bh + 1) * smem_w + (tx + 1)] = (r + 1 < height && c >= 0 && c < width) ? in[(r+1)*width + c] : 0;
    if (tx == 0)
```

```
    smem[(ty + 1) * smem_w + 0] = (r >= 0 && r < height && c - 1 >= 0) ? in[r*width + (c-1)] : 0;
  if (tx == bw - 1)
    smem[(ty + 1) * smem_w + (bw + 1)] = (r >= 0 && r < height && c + 1 < width) ? in[r*width + (c+1)] :
0;


  if (tx == 0 && ty == 0)
    smem[0 * smem_w + 0] = (r - 1 >= 0 && c - 1 >= 0) ? in[(r-1)*width + (c-1)] : 0;
  if (tx == bw - 1 && ty == 0)
    smem[0 * smem_w + (bw + 1)] = (r - 1 >= 0 && c + 1 < width) ? in[(r-1)*width + (c+1)] : 0;
  if (tx == 0 && ty == bh - 1)
    smem[(bh + 1) * smem_w + 0] = (r + 1 < height && c - 1 >= 0) ? in[(r+1)*width + (c-1)] : 0;
  if (tx == bw - 1 && ty == bh - 1)
    smem[(bh + 1) * smem_w + (bw + 1)] = (r + 1 < height && c + 1 < width) ? in[(r+1)*width + (c+1)] :
0;


  __syncthreads();


  if (r >= height || c >= width) return;
  if (r == 0 || r == height - 1 || c == 0 || c == width - 1) {
    out[r*width + c] = 0;
    return;
  }


  int sr = ty + 1;
  int sc = tx + 1;
  int p1 = smem[(sr - 1) * smem_w + (sc - 1)];
  int p2 = smem[(sr - 1) * smem_w + sc];
  int p3 = smem[(sr - 1) * smem_w + (sc + 1)];
  int q1 = smem[sr * smem_w + (sc - 1)];
```

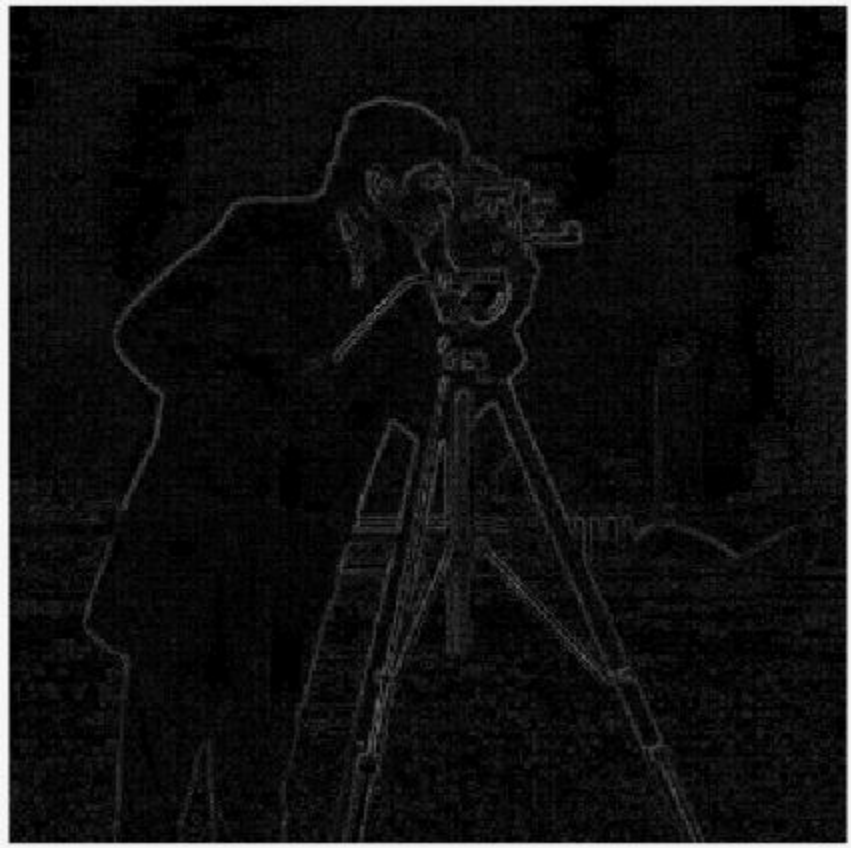
```
    int q3 = smem[sr * smem_w + (sc + 1)];
    int r1 = smem[(sr + 1) * smem_w + (sc - 1)];
    int r2 = smem[(sr + 1) * smem_w + sc];
    int r3 = smem[(sr + 1) * smem_w + (sc + 1)];

    int qx = (p1 - r1) + 2*(p2 - r2) + (p3 - r3);
    int qy = (p1 - p3) + 2*(q1 - q3) + (r1 - r3);
    int mag = abs(qx) + abs(qy);
    if (mag > 255) mag = 255;

    out[r*width + c] = (uint8_t)mag;
}
```

2.2 Part 2 Output Images/Execution Time

Configuration 1 Execution:

```
==PROF== Disconnected from process 1956256
[1956256] sobel_cuda_shared@127.0.0.1
  sobel_1d_shared(const unsigned char *, unsigned char *, int, int) (25000, 1, 1)x(1000, 1, 1), Device 0, CC 8.9, Invocations 1
    Section: GPU Speed Of Light Throughput
    ----------------------- ----------- ------------ ------------ ------------
    Metric Name             Metric Unit      Minimum      Maximum      Average
    ----------------------- ----------- ------------ ------------ ------------
    DRAM Frequency                  Ghz         6.24         6.24         6.24
    SM Frequency                    Mhz       794.94       794.94       794.94
    Elapsed Cycles                cycle 1,025,011.00 1,025,011.00 1,025,011.00
    Memory Throughput                 %        44.14        44.14        44.14
    DRAM Throughput                   %         9.69         9.69         9.69
    Duration                         ms         1.29         1.29         1.29
    L1/TEX Cache Throughput           %        48.09        48.09        48.09
    L2 Cache Throughput               %         8.73         8.73         8.73
    SM Active Cycles              cycle   939,789.29   939,789.29   939,789.29
    Compute (SM) Throughput           %        44.14        44.14        44.14
    ----------------------- ----------- ------------ ------------ ------------

    Section: GPU and Memory Workload Distribution
    -------------------------- ----------- -------------- -------------- --------------
    Metric Name                Metric Unit        Minimum        Maximum        Average
    -------------------------- ----------- -------------- -------------- --------------
    Average DRAM Active Cycles       cycle     779,965.33     779,965.33     779,965.33
    Total DRAM Elapsed Cycles        cycle  48,307,200.00  48,307,200.00  48,307,200.00
    Average L1 Active Cycles         cycle     939,789.29     939,789.29     939,789.29
    Total L1 Elapsed Cycles          cycle  59,389,474.00  59,389,474.00  59,389,474.00
    Average L2 Active Cycles         cycle     971,064.83     971,064.83     971,064.83
    Total L2 Elapsed Cycles          cycle  27,399,432.00  27,399,432.00  27,399,432.00
    Average SM Active Cycles         cycle     939,789.29     939,789.29     939,789.29
    Total SM Elapsed Cycles          cycle  59,389,474.00  59,389,474.00  59,389,474.00
    Average SMSP Active Cycles       cycle     927,007.37     927,007.37     927,007.37
    Total SMSP Elapsed Cycles        cycle 237,557,896.00 237,557,896.00 237,557,896.00
    -------------------------- ----------- -------------- -------------- --------------

    Section: Launch Statistics
    -------------------------------- --------------- ------------- ------------- -------------
    Metric Name                          Metric Unit       Minimum       Maximum       Average
    -------------------------------- --------------- ------------- ------------- -------------
    Block Size                                            1,000.00      1,000.00      1,000.00
    Grid Size                                            25,000.00     25,000.00     25,000.00
    Registers Per Thread             register/thread         20.00         20.00         20.00
    Shared Memory Configuration Size           Kbyte          8.19          8.19          8.19
    Driver Shared Memory Per Block       Kbyte/block          1.02          1.02          1.02
    Dynamic Shared Memory Per Block      Kbyte/block          3.01          3.01          3.01
    Static Shared Memory Per Block        byte/block          0.00          0.00          0.00
    # SMs                                         SM         58.00         58.00         58.00
    Stack Size                                            1,024.00      1,024.00      1,024.00
    Threads                                     thread 25,000,000.00 25,000,000.00 25,000,000.00
    # TPCs                                               29.00         29.00         29.00
    Uses Green Context                                     0.00          0.00          0.00
    Waves Per SM                                         431.03        431.03        431.03
    -------------------------------- --------------- ------------- ------------- -------------

    Section: Occupancy
    ------------------------------ ----------- ------- ------- -------
    Metric Name                    Metric Unit Minimum Maximum Average
    ------------------------------ ----------- ------- ------- -------
    Block Limit SM                       block   24.00   24.00   24.00
    Block Limit Registers                block    2.00    2.00    2.00
    Block Limit Shared Mem               block    2.00    2.00    2.00
    Block Limit Warps                    block    1.00    1.00    1.00
    Theoretical Active Warps per SM       warp   32.00   32.00   32.00
    Theoretical Occupancy                    %   66.67   66.67   66.67
    Achieved Occupancy                       %   64.60   64.60   64.60
    Achieved Active Warps Per SM          warp   31.01   31.01   31.01
    ------------------------------ ----------- ------- ------- -------

  Note: The shown averages are calculated as the arithmetic mean of the metric values after the evaluation of the
  metrics for each individual kernel launch.
  If aggregating across varying launch configurations (like shared memory, cache config settings), the arithmetic
  mean can be misleading and looking at the individual results is recommended instead.
  This output mode is backwards compatible to the per-kernel summary output of nvprof

  grid=(25000,1,1)  block=(1000,1,1)


Total program wall-clock time: 2.06248081 sec
```

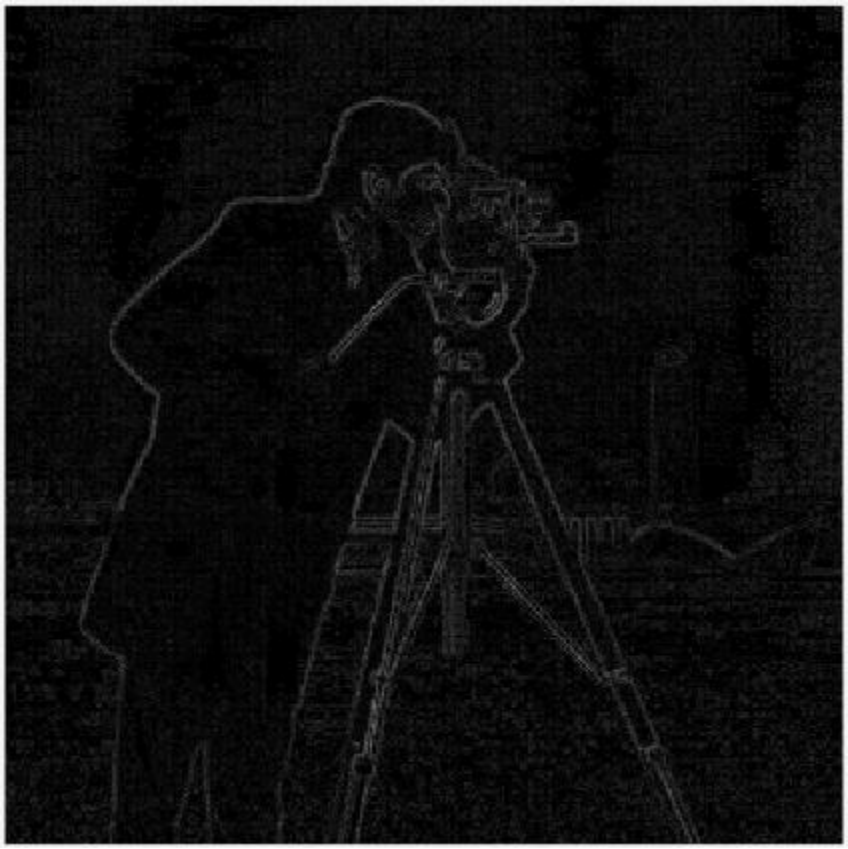
Configuration 1 Output:



Configuration 2 Execution:

```
==PROF== Disconnected from process 3072460
[3072460] sobel_cuda_shared@127.0.0.1
  sobel_2d_shared(const unsigned char *, unsigned char *, int, int) (50, 500, 1)x(100, 10, 1), Device 0, CC 8.9, Invocations 1
    Section: GPU Speed Of Light Throughput
    ----------------------- ----------- ------------ ------------ ------------
    Metric Name             Metric Unit      Minimum      Maximum      Average
    ----------------------- ----------- ------------ ------------ ------------
    DRAM Frequency                  Ghz         6.24         6.24         6.24
    SM Frequency                    Mhz       794.95       794.95       794.95
    Elapsed Cycles                cycle 1,169,964.00 1,169,964.00 1,169,964.00
    Memory Throughput                 %        35.39        35.39        35.39
    DRAM Throughput                   %         9.02         9.02         9.02
    Duration                         ms         1.47         1.47         1.47
    L1/TEX Cache Throughput           %        38.26        38.26        38.26
    L2 Cache Throughput               %        10.38        10.38        10.38
    SM Active Cycles              cycle 1,081,129.10 1,081,129.10 1,081,129.10
    Compute (SM) Throughput           %        35.39        35.39        35.39
    ----------------------- ----------- ------------ ------------ ------------

    Section: GPU and Memory Workload Distribution
    ------------------------- ----------- -------------- -------------- --------------
    Metric Name               Metric Unit        Minimum        Maximum        Average
    ------------------------- ----------- -------------- -------------- --------------
    Average DRAM Active Cycles       cycle     829,378.67     829,378.67     829,378.67
    Total DRAM Elapsed Cycles        cycle  55,139,328.00  55,139,328.00  55,139,328.00
    Average L1 Active Cycles         cycle   1,081,129.10   1,081,129.10   1,081,129.10
    Total L1 Elapsed Cycles          cycle  67,778,678.00  67,778,678.00  67,778,678.00
    Average L2 Active Cycles         cycle   1,029,448.67   1,029,448.67   1,029,448.67
    Total L2 Elapsed Cycles          cycle  29,391,960.00  29,391,960.00  29,391,960.00
    Average SM Active Cycles         cycle   1,081,129.10   1,081,129.10   1,081,129.10
    Total SM Elapsed Cycles          cycle  67,778,678.00  67,778,678.00  67,778,678.00
    Average SMSP Active Cycles       cycle   1,067,359.89   1,067,359.89   1,067,359.89
    Total SMSP Elapsed Cycles        cycle 271,114,712.00 271,114,712.00 271,114,712.00
    ------------------------- ----------- -------------- -------------- --------------

    Section: Launch Statistics
    -------------------------------- --------------- ------------- ------------- -------------
    Metric Name                          Metric Unit       Minimum       Maximum       Average
    -------------------------------- --------------- ------------- ------------- -------------
    Block Size                                             1,000.00      1,000.00      1,000.00
    Grid Size                                             25,000.00     25,000.00     25,000.00
    Registers Per Thread             register/thread         23.00         23.00         23.00
    Shared Memory Configuration Size           Kbyte          8.19          8.19          8.19
    Driver Shared Memory Per Block       Kbyte/block          1.02          1.02          1.02
    Dynamic Shared Memory Per Block      Kbyte/block          1.22          1.22          1.22
    Static Shared Memory Per Block        byte/block          0.00          0.00          0.00
    # SMs                                         SM         58.00         58.00         58.00
    Stack Size                                            1,024.00      1,024.00      1,024.00
    Threads                                     thread 25,000,000.00 25,000,000.00 25,000,000.00
    # TPCs                                               29.00         29.00         29.00
    Uses Green Context                                     0.00          0.00          0.00
    Waves Per SM                                         431.03        431.03        431.03
    -------------------------------- --------------- ------------- ------------- -------------

    Section: Occupancy
    ------------------------------ ----------- ------- ------- -------
    Metric Name                    Metric Unit Minimum Maximum Average
    ------------------------------ ----------- ------- ------- -------
    Block Limit SM                       block   24.00   24.00   24.00
    Block Limit Registers                block    2.00    2.00    2.00
    Block Limit Shared Mem               block    3.00    3.00    3.00
    Block Limit Warps                    block    1.00    1.00    1.00
    Theoretical Active Warps per SM       warp   32.00   32.00   32.00
    Theoretical Occupancy                    %   66.67   66.67   66.67
    Achieved Occupancy                       %   64.78   64.78   64.78
    Achieved Active Warps Per SM          warp   31.09   31.09   31.09
    ------------------------------ ----------- ------- ------- -------

  Note: The shown averages are calculated as the arithmetic mean of the metric values after the evaluation of the
  metrics for each individual kernel launch.
  If aggregating across varying launch configurations (like shared memory, cache config settings), the arithmetic
  mean can be misleading and looking at the individual results is recommended instead.
  This output mode is backwards compatible to the per-kernel summary output of nvprof

  grid=(50,500,1)  block=(100,10,1)


Total program wall-clock time: 1.97395708 sec

[tbarber@login8 Part2]$
```

Configuration 2:



2.3 Performance Table

Grid Configuration	Block Configuration	Kernel Time (s)	Wall Time (s)
(25000,1,1)	(1000,1,1)	0.00129	2.062
(50,500,1)	(100,10,1)	0.00147	1.974

2.4 Comparison with Lab 4

Implementation	Execution Time (s)
CPU	2.037
CPU + MPI	1.014
GPU (Shared Memory)	1.974

2.5 Analysis

The shared memory implementation was expected to improve performance due to reduced global memory access latency. However, the results indicate that shared memory did not outperform the global memory implementation in terms of overall wall clock time.

Although kernel execution time improved in some configurations, the additional overhead associated with loading data into shared memory and thread synchronization reduced the overall benefit.

The best configuration for shared memory was:

Grid = (50,500,1), Block = (100,10,1)

Global vs Shared Memory Comparison

3.1 Performance Summary

Implementation	Execution Time (s)
CPU	2.037
CPU + MPI	1.014
GPU (Global Memory)	1.815
GPU (Shared Memory)	1.974

3.2 Discussion

Although GPUs provide massive parallelism, the MPI implementation achieved the best performance in this lab. This is likely due to the relatively simple computation of the Sobel operator and the overhead associated with GPU memory transfers and kernel launches.

Shared memory did not provide a performance advantage in this implementation, as the cost of loading data into shared memory and synchronizing threads outweighed its benefits.

The global memory implementation performed better due to simpler memory access patterns and lower overhead.

This demonstrates that performance depends not only on hardware capability but also on how well the algorithm maps to the architecture.

---

4. Conclusion

In this lab, CUDA-based implementations of the Sobel edge detection algorithm were developed using both global and shared memory.

Key findings:

- MPI implementation achieved the best performance in this lab

- GPU implementations did not outperform MPI due to overhead costs
- Global memory performed better than shared memory in this case
- Proper grid and block configuration is critical for performance